

EXPLORATORY DATA ANALYSIS

Optimising NYC Taxi Operations

Assignment ID: EDA/02
Business Analytics / Exploratory Data Analysis Report

Student Name: Varun Gahlot
upGrad
2023 NYC Yellow Taxi Data

1. Executive Summary

This report documents the completed exploratory data analysis of 2023 New York City yellow taxi trip data. The analysis follows the assignment structure: data preparation and cleaning, general EDA, detailed operational analysis, and business recommendations. The central objective is to understand where and when taxi demand is concentrated and how those patterns can be used to improve routing, dispatching, fleet positioning and pricing.

The analysis shows a strong time-and-location dependence in demand. In the sampled data, 18:00 is the busiest hour overall, with 20,222 trips; after scaling by the sampling ratio, the five busiest hours correspond to approximately 202,220 trips at 18:00, 192,040 at 17:00, 179,800 at 19:00, 176,270 at 16:00 and 176,080 at 15:00. Weekday demand peaks at 18:00 with 15,412 sampled trips, while weekend demand peaks at 17:00 with 4,835.

Revenue is also concentrated in daytime hours: 87.59% of analysed revenue is attributed to daytime trips and 12.41% to nighttime trips. Shorter journeys have substantially higher fare-per-mile values, and vendor differences are most pronounced for trips up to two miles. Tipping is strongest for short trips and weakest for longer trips. Weekend trips have slightly higher average passenger counts than weekday trips.

Taken together, the findings support a demand-responsive operating strategy: proactively position cabs before peak periods, prioritise high-demand zones, treat night demand as a separate geographic pattern, and use distance-tiered competitive pricing rather than a single fare-per-mile benchmark.

2. Business Problem and Objectives

The assignment asks the analyst to use exploratory analysis to identify operational inefficiencies and actionable patterns in NYC taxi operations. The report therefore focuses on demand, geography, travel efficiency, revenue, pricing, customer tipping, occupancy and surcharges.

- Identify important temporal and geographic demand drivers.
- Measure trip, fare and revenue behaviour using derived metrics.
- Identify slow routes and high-demand pickup/drop-off locations.
- Compare weekday and weekend demand and day versus night revenue.
- Compare vendor fare-per-mile behaviour across hours and distance tiers.
- Understand tipping and passenger-count patterns.
- Translate the analysis into routing, dispatching, positioning and pricing recommendations.

3. Data Preparation and Cleaning

3.1 Data preparation

The notebook combines the supplied 2023 taxi data and derives analytical fields including pickup hour, pickup date, pickup day, pickup month and trip duration. The analysis is based on the sampled dataset used in the notebook; the assignment explicitly requires scaling sampled trip counts to estimate actual trip volumes.

3.2 Monetary-value checks

The notebook checked for negative monetary values and found no negative monetary observations in the relevant fields. Zero values were also examined: fare_amount had 98 zero values, tip_amount 65,409, total_amount 46 and trip_distance 5,625. Zero trip distance was not automatically treated as invalid because trips can begin and end in the same zone.

3.3 Missing values

Before imputation, missingness was concentrated in passenger_count, RatecodeID, store_and_fwd_flag, congestion_surcharge and airport_fee. The notebook subsequently completed the missing-value handling and confirmed zero missing values across the analysed dataframe.

3.4 Outlier treatment

The notebook investigated records such as near-zero distance with unusually high fares, zero distance and zero fare with different pickup/drop-off zones, trips above 250 miles, and invalid payment type 0. It found 6 near-zero-distance high-fare cases, 7 zero-distance/zero-fare different-zone cases, 3 trips above 250 miles and 9,432 payment-type-0 records at the outlier-check stage. After cleaning, payment type 0, passenger counts above 6 and trip distances above 250 miles were absent from the final EDA dataset.

4. General Exploratory Data Analysis

4.1 Variable classification

Numerical variables include passenger_count, trip_distance, fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, total_amount, congestion_surcharge, airport_fee and derived numerical measures such as trip duration. Categorical/discrete identifiers include VendorID, RatecodeID, PULocationID, DOLocationID, payment_type and store_and_fwd_flag; datetime fields are treated as temporal variables.

4.2 Financial and trip validation

After the cleaning steps, the main EDA dataset contained 275,234 rows for several of the validation checks, with payment type 0, passenger counts above 6 and trip distances above 250 miles removed. A subsequent cleaned dataset used for payment-type analysis contained 284,663 rows after targeted invalid-value filtering. The exact row count therefore varies by analysis because different checks apply different filters.

4.3 Distance and fare relationship

The notebook calculated a correlation of 0.0896 between trip_distance and fare_amount in the filtered dataset. This is a weak positive linear correlation in the sampled/cleaned data, indicating that distance alone does not fully explain fare variation; pricing components, trip characteristics and other charges also matter.

4.4 Other correlations

Variables	Correlation	Interpretation
Fare vs trip duration	0.2711	Weak-to-moderate positive relationship
Fare vs passenger count	0.0435	Very weak relationship
Tip vs trip distance	0.5753	Moderate positive relationship

The strongest of these three correlations is between tip amount and trip distance. Passenger count has almost no linear relationship with fare amount in the analysed data.

4.5 Payment type distribution

Payment type	Trips
1	233,637
2	47,690
3	1,298
4	2,038

Payment type 1 is dominant in the cleaned payment-type analysis, accounting for the large majority of trips. This concentration should be considered when designing payment-related customer or operational interventions.

5. Geographic Analysis

The taxi-zone shapefile was loaded and merged with trip data using LocationID and PULocationID. The resulting geographic analysis enabled zone-level trip counts and a colour-coded zone map in the notebook.

Rank	Pickup Zone	Trip count	Zone name / context
1	132	14,568	JFK Airport
2	237	13,235	Upper East Side South
3	161	13,139	Midtown Center
4	236	11,954	Upper East Side North
5	162	10,138	Midtown East
6	138	9,609	LaGuardia Airport
7	186	9,534	Penn Station/Madison Sq West
8	230	9,411	Times Sq/Theatre District
9	142	9,290	Lincoln Square East
10	170	8,350	Murray Hill

The concentration of trips in airport and central Manhattan zones demonstrates that geography is a major driver of taxi demand. These zones are therefore natural candidates for proactive positioning and demand forecasting.

6. Detailed EDA: Operational Efficiency and Demand

6.1 Slow routes

The route-speed analysis grouped trips by pickup zone, drop-off zone and pickup hour and calculated average distance, average duration and average speed. The slowest observed combinations included 226→145 at 18:00 (average speed approximately 0.027 miles/minute), 113→113 at 13:00 (0.038), and 260→129 at 17:00 (0.041). These extreme values have very low trip counts in several cases, so they should be treated as indicators for investigation rather than as standalone evidence of systemic congestion.

6.2 Hourly demand

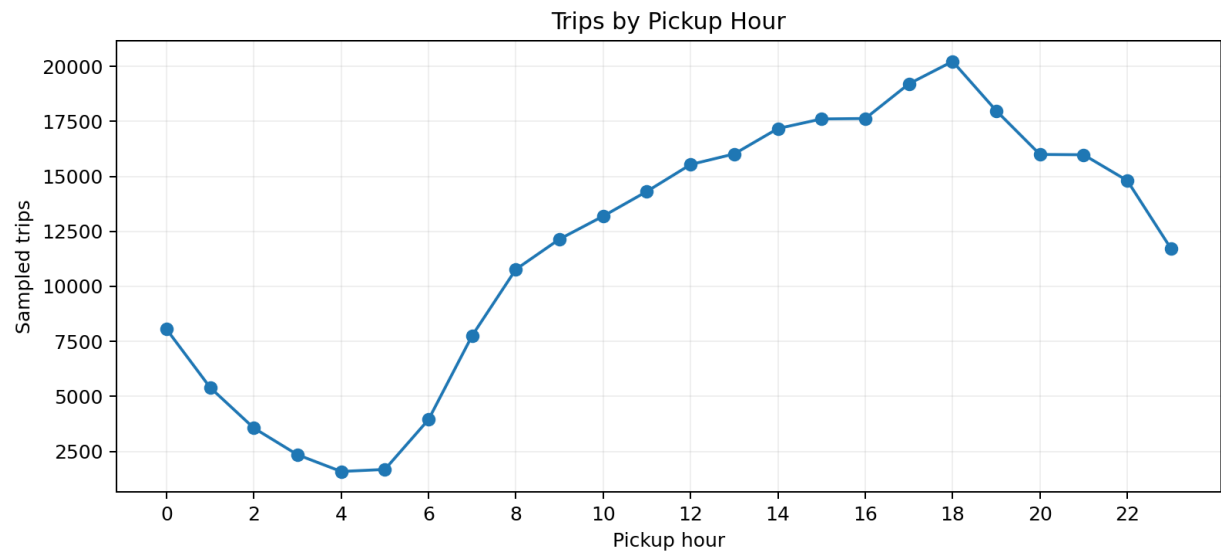


Figure 1. Sampled trips by pickup hour.

Hour	Sampled trips
15:00	17,608
16:00	17,627
17:00	19,204
18:00	20,222
19:00	17,980

Demand rises strongly from the morning into the afternoon and reaches its maximum at 18:00. The five busiest sampled hours are 18:00, 17:00, 19:00, 16:00 and 15:00.

6.3 Scaled trip volume

Hour	Estimated actual trips
18:00	202,220
17:00	192,040
19:00	179,800
16:00	176,270
15:00	176,080

These values are scaled estimates based on the notebook's sampling ratio. They should be interpreted as estimates of the underlying 2023 trip volume rather than exact counts.

6.4 Weekday versus weekend

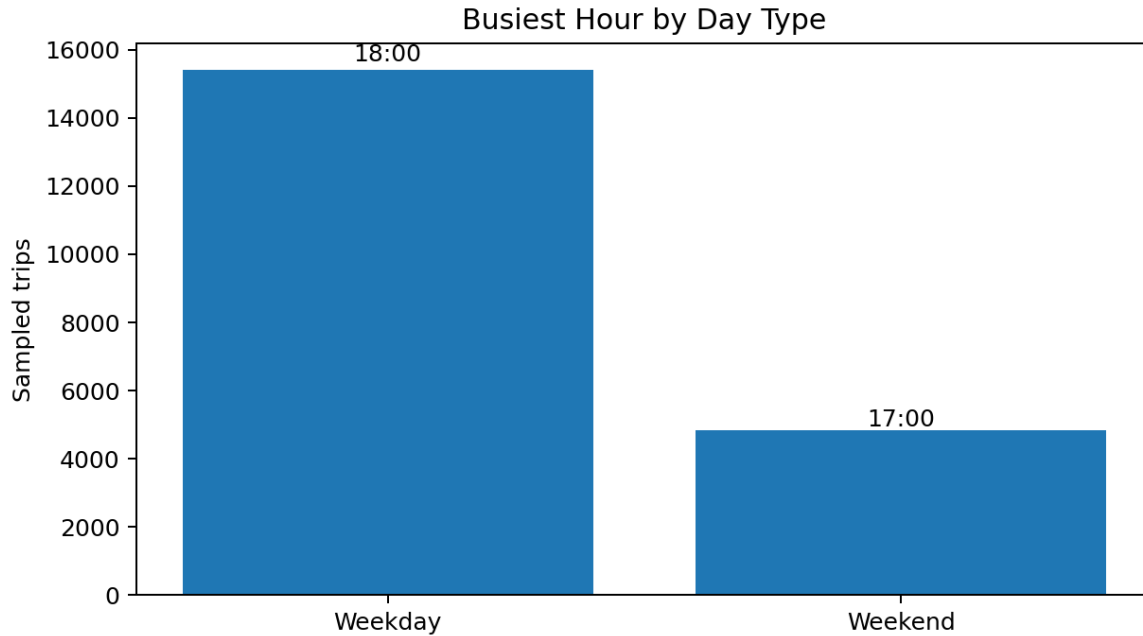


Figure 2. Busiest observed hour for weekdays and weekends.

The weekday peak occurs at 18:00 with 15,412 sampled trips, while the weekend peak occurs at 17:00 with 4,835. Operational schedules should therefore distinguish weekday and weekend demand rather than applying one static peak-hour plan.

7. Pickup/Drop-off Zone Behaviour

Rank	Top pickup zone	Pickups	Top drop-off zone	Drop-offs
1	132	14,568	236	12,640
2	237	13,235	237	11,861
3	161	13,139	161	11,080
4	236	11,954	230	8,610
5	162	10,138	170	8,419
6	138	9,609	162	8,036
7	186	9,534	142	8,014
8	230	9,411	239	7,884
9	142	9,290	141	7,360
10	170	8,350	68	7,283

7.1 Pickup/drop-off imbalance

Zone 70 has the highest pickup/drop-off ratio at 7.81, followed by Zone 132 at 4.34 and Zone 138 at 2.63. These ratios indicate zones where pickups substantially exceed drop-offs and may therefore create vehicle-availability or repositioning requirements. Several zones have a ratio of zero because they recorded drop-offs but no pickups in the sample.

7.2 Night demand

Rank	Night pickup zone	Pickups	Night drop-off zone	Drop-offs
1	79	2,478	79	1,420
2	132	2,199	48	1,178
3	249	2,015	170	1,038
4	48	1,593	68	1,002

5	148	1,548	107	964
6	114	1,341	141	907
7	230	1,330	263	903
8	186	1,072	236	819
9	164	962	249	793
10	68	945	230	781

Night demand shifts toward a distinct set of zones. Zones 79, 132, 249, 48 and 148 are especially important for nighttime pickup positioning.

8. Revenue and Pricing Analysis

8.1 Day versus night revenue

Period	Revenue	Share
Day	\$7,196,305.45	87.59%
Night	\$1,019,946.97	12.41%

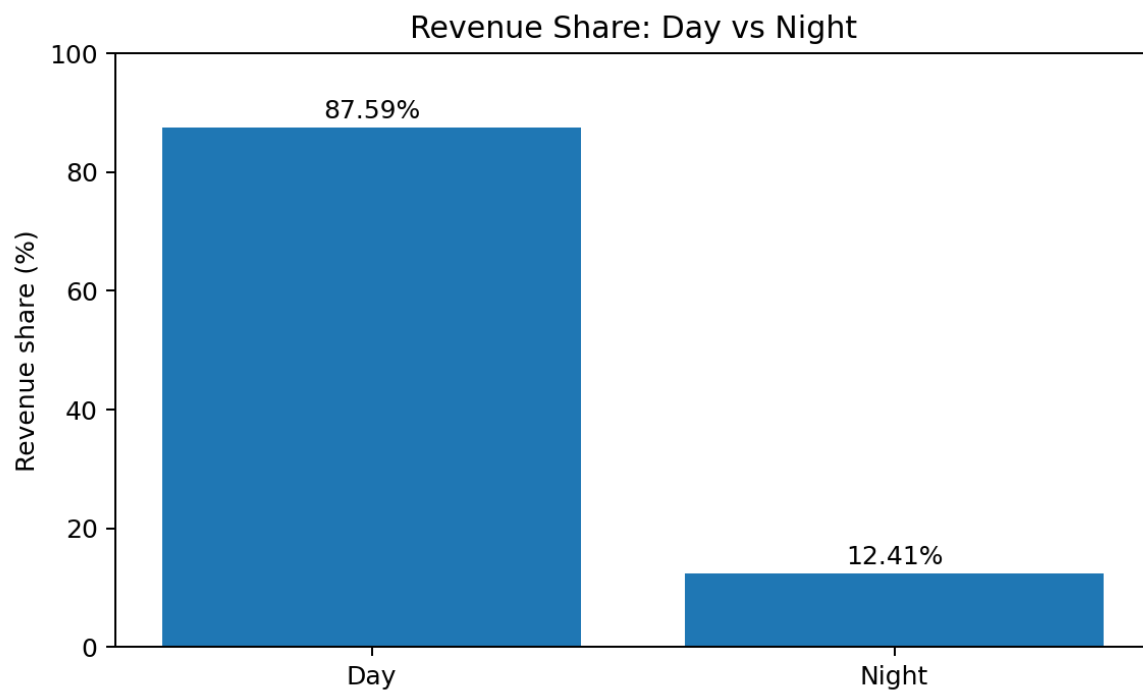


Figure 3. Revenue share by time period.

Daytime trips contribute the overwhelming majority of analysed revenue. The implication is not simply to reduce night supply, but to target night supply to the zones where demand remains concentrated.

8.2 Fare per mile by passenger count

Passengers	Average fare per mile per passenger
1	10.74
2	6.31
3	4.29
4	4.98
5	1.58
6	1.28

Fare per mile per passenger generally decreases as passenger count rises, because the trip economics are distributed across more passengers. The four-passenger group is a small non-monotonic exception.

8.3 Fare per mile by hour and day

Metric	Highest	Lowest
Hour	04:00 — 18.24	07:00 — 8.55
Day	Thursday — 12.43	Friday — 9.91

The hourly fare-per-mile profile is highly variable. The unusually high 04:00 value should be interpreted cautiously because it may be influenced by the composition and volume of trips at that hour.

8.4 Vendor comparison by distance tier

Distance tier	Vendor 1	Vendor 2	Vendor 6
0–2 miles	9.76	17.92	30.16
2–5 miles	6.38	6.53	7.00
>5 miles	4.41	4.49	4.05

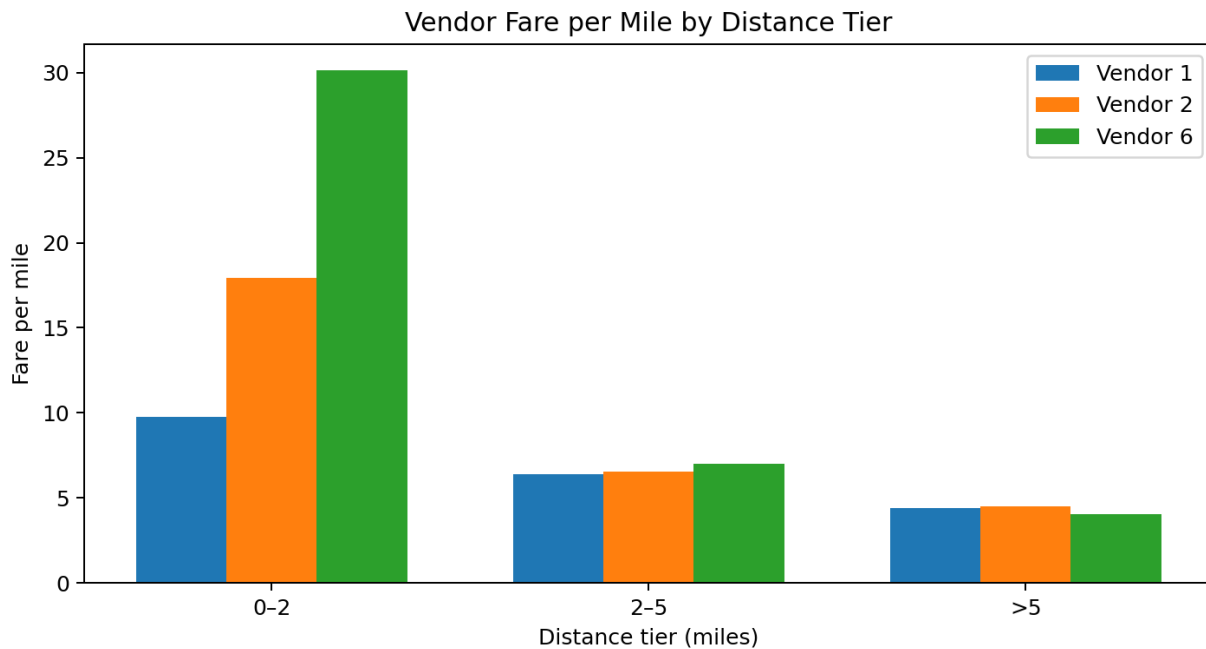


Figure 4. Vendor fare-per-mile comparison by distance tier.

Vendor differences are strongest for short trips. Vendor 6 has the highest short-trip average fare per mile, followed by Vendor 2 and Vendor 1. In the 2–5 mile tier the values are close, and above 5 miles they converge further.

9. Customer Experience: Tips

Distance tier	Average tip %
0–2 miles	22.44%
2–5 miles	18.61%
>5 miles	16.45%

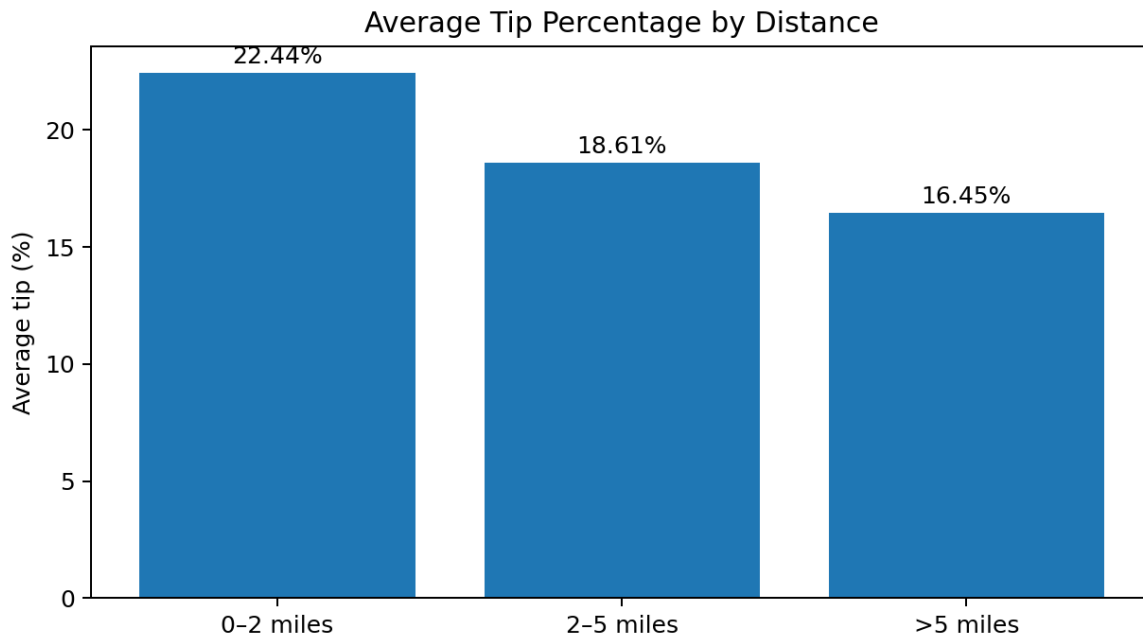


Figure 5. Average tip percentage by distance tier.

Short trips generate the highest average tip percentage, while long trips generate the lowest. Passenger count also shows variation, with four-passenger trips recording the lowest average tip percentage at 17.42%.

Tip percentage also varies by pickup hour. It is lowest around 05:00 (16.96%) and highest around 18:00 (22.07%). When low-tip trips (<10%) are compared with high-tip trips (>25%), average distance is 3.92 miles versus 2.30 miles, while average passenger counts and pickup hours are similar. Distance is therefore the clearest factor associated with higher tipping in this analysis.

10. Passenger Count Analysis

10.1 By hour and day

Measure	Highest	Lowest
Hour	01:00 — 1.42	06:00 — 1.21
Day	Saturday — 1.44	Tuesday/Wednesday — 1.30

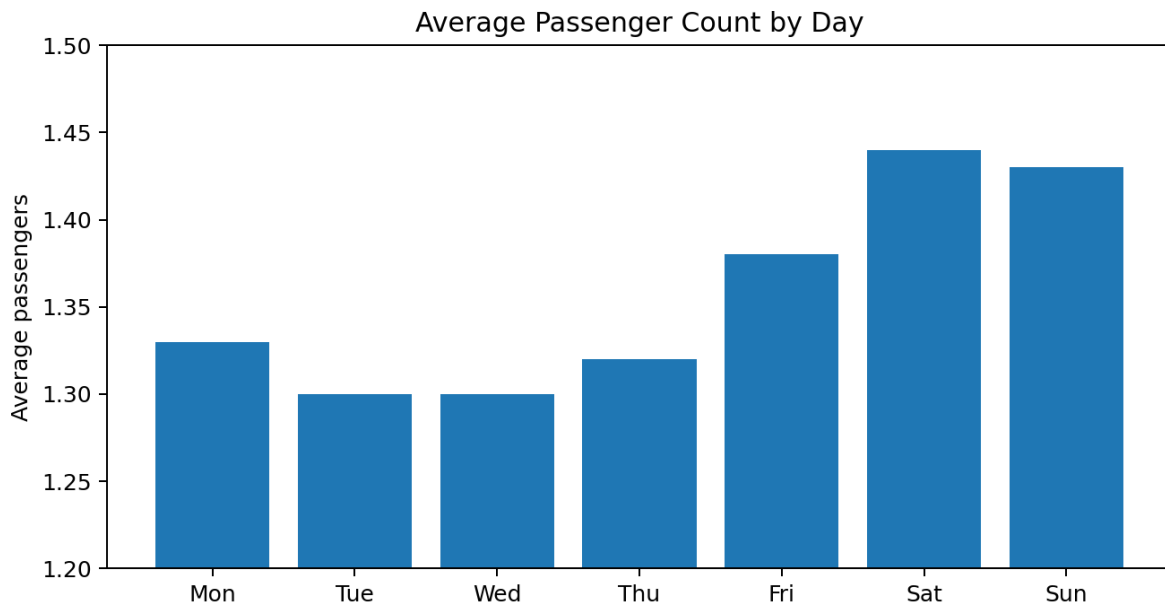


Figure 6. Average passenger count by day of week.

Average occupancy is lowest in the early morning and generally higher late at night. Weekends have higher average passenger counts than weekdays, with Saturday at 1.44 and Sunday at 1.43.

10.2 By pickup zone

High-average zones	Average passengers	Low-average zones	Average passengers
6	2.00	242	1.00
120	2.00	241	1.00
253	2.00	258	1.00
154	2.00	254	1.00
67	1.80	167	0.92
175	1.71	32	0.88
12	1.71	26	0.75
195	1.68	109	0.50

Passenger occupancy varies by pickup zone. Zones with higher occupancy may be suitable for targeted supply planning, although average passenger count should be interpreted together with trip volume rather than in isolation.

11. Surcharge and Extra-Charge Analysis

Charge field	Frequency
Improvement surcharge	284,337
MTA tax	281,838
Congestion surcharge	253,981
Extra	171,051
Tolls amount	22,976
Airport fee	1,958

The notebook found 284,353 trips with at least one surcharge and 118 trips without a surcharge. Surcharge frequency is therefore extremely high in the analysed dataset.

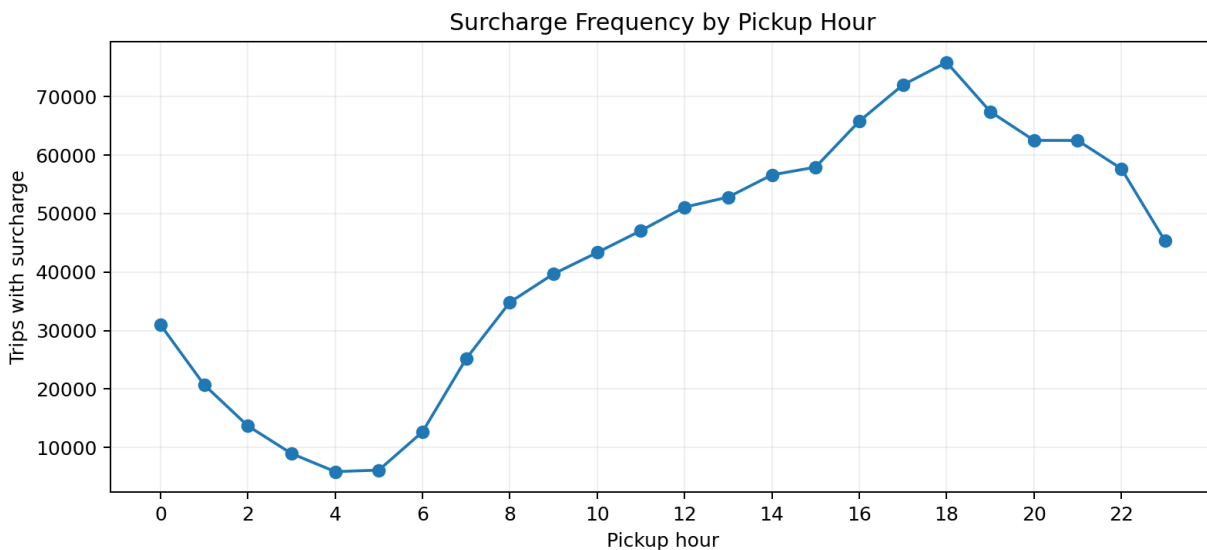


Figure 7. Surcharge frequency by pickup hour.

Surcharge frequency rises sharply through the day and peaks around the evening commute. In the updated notebook, 18:00 records 75,883 surcharge-associated trips, followed by 17:00 with 72,016 and 19:00 with 67,444. Zone 132 is the leading pickup zone with 51,655 surcharge-associated trips, while Zone 236 leads drop-offs with 45,165.

12. Final Insights and Recommendations

12.1 Routing and dispatching

- Increase cab availability before the 15:00–19:00 demand build-up, with special attention to the 18:00 weekday peak.
- Prioritise high-demand pickup zones such as 132, 161, 237, 236 and 138 during major daytime/early-evening peaks.
- Use a separate night dispatch profile focused on Zones 79, 132, 249, 48 and 148.
- Use time-and-zone demand forecasts to reposition idle cabs proactively and reduce empty travel.
- Flag extreme slow-route observations for further investigation, but validate them using larger trip counts before making permanent routing changes.

12.2 Strategic cab positioning

- Position additional supply in the strongest pickup zones ahead of the evening peak rather than reacting only after queues appear.
- Shift the geographic mix of available cabs after 23:00 toward the high-night-demand zones.
- Plan slightly higher weekend occupancy capacity because Saturday and Sunday have the highest average passenger counts.
- Use zone-level pickup/drop-off ratios to identify locations where vehicles may need proactive repositioning.
- Combine historical patterns with real-time demand signals for dynamic positioning.

12.3 Pricing strategy

- Benchmark short-trip fares carefully because vendor differences are largest in the 0–2 mile tier.
- Maintain competitive rates in the 2–5 mile segment where vendor fare-per-mile values are relatively close.
- Use distance-tiered pricing analysis rather than a single overall fare-per-mile benchmark.
- Consider demand, time, distance and competitive context together when evaluating pricing changes.
- Use tip behaviour as a customer-experience signal: short trips have higher average tip percentages, while longer trips have lower percentages.

13. Key Business Takeaways

Business question	Key answer
When is demand highest?	18:00 overall; weekday peak 18:00; weekend peak 17:00.
What are the five busiest hours?	18:00, 17:00, 19:00, 16:00, 15:00.
Where is demand concentrated?	Airport and central Manhattan zones dominate overall pickups.
Where is night demand concentrated?	Zones 79, 132, 249, 48 and 148 are key nighttime pickup areas.
Day vs night revenue?	87.59% day; 12.41% night.
How does fare/mile change with distance?	It declines substantially as distance increases.
Where are vendor differences strongest?	0–2 mile trips.
What drives higher tip percentage?	Shorter distance is the clearest factor in this analysis.
When are passenger counts highest?	Saturday/Sunday and later hours.
When are surcharges most frequent?	Afternoon/evening, peaking around 18:00.

14. Assumptions and Limitations

- The trip counts used for operational scaling are estimates because the notebook analyses a sample rather than the full population.
- Location IDs are treated as the geographic operating units for zone-level decisions.
- Zero trip distance is not automatically invalid because pickup and drop-off may occur in the same zone.
- Extreme route-duration/speed observations with very low trip counts should not be interpreted as representative congestion without further validation.
- Fare-per-mile comparisons are descriptive and do not establish causality or prove that one vendor is universally more expensive.
- Tip analysis is based on recorded tip amounts and therefore reflects the limitations of the underlying payment/tip recording process.
- Recommendations are derived from observed patterns and should be validated with real-time demand, driver availability, traffic conditions and competitive pricing before operational deployment.

15. Conclusion

The completed analysis tells a consistent operational story. NYC taxi demand is highly time- and location-dependent, with the strongest overall activity concentrated in the late afternoon and early evening. The 18:00 weekday peak is the clearest point at which additional supply and proactive

dispatching can create value. At the same time, nighttime demand is geographically distinct, so night operations should not simply mirror the daytime positioning strategy.

Pricing behaviour is also segmented. Short trips have higher fare-per-mile values and higher average tip percentages, while long trips have lower fare-per-mile and tip percentages. Vendor differences are most visible for short journeys, making competitive benchmarking especially important in the 0–2 mile segment.

The recommended operating model is therefore dynamic: position cabs where demand is expected, reposition them as demand moves, use day-of-week and time-of-day profiles, and evaluate pricing by distance tier and competitive context. This approach aligns fleet supply with observed demand, reduces avoidable idle time and supports revenue optimisation while maintaining a customer-focused service.

Appendix A. Notebook Outputs Used in This Report

This report was prepared from the updated completed notebook uploaded for the assignment.

Numerical values in this report are taken from the notebook outputs; charts in the report reproduce the key numerical findings for readability.

Assignment report checklist:

- Data preparation and cleaning covered.
- Missing-value and outlier handling documented.
- General EDA and correlations documented.
- Geographic zone analysis documented.
- Operational efficiency and hourly demand documented.
- Weekday/weekend and night demand documented.
- Revenue share and pricing analysis documented.
- Vendor and distance-tier analysis documented.
- Tip and passenger-count analysis documented.
- Surcharge analysis documented.
- Routing, positioning and pricing recommendations documented.
- Assumptions and limitations documented.